

Adam Mrotek

Full-Stack & AI Engineer

London, United Kingdom (Hybrid / Remote) · adam.mrotek@gmail.com · LinkedIn · GitHub · Portfolio

PROFESSIONAL EXPERIENCE

Front-end Engineer | Croud Jul 2023 – Mar 2026
London, UK (Hybrid)

- **Full-Stack Product Delivery:** Operate within a high-leverage 3-person team building Croud's core marketing-platform suite, driving solo ownership of features from technical roadmap to containerized production UI.
- **Croud Control (3,000+ Users):** Engineered core product features across agency and client divisions. Spearheaded frontend implementation with strict test coverage and migrated legacy Nuxt architectures to modern TypeScript & Vue 3.
- **Creator Platform Build:** Delivered a pixel-perfect, highly responsive influencer-marketing frontend against complex Figma specifications. Authored all supporting FastAPI endpoints to build a robust internal alternative to a multi-million-pound external agency proposal, successfully retaining a key client opportunity.
- **Taxonomy Tool:** Unblocked a stalled internal product by single-handedly driving full vertical vertical-slice development, encompassing PostgreSQL database schema changes, FastAPI backend validation, and the production UI layer.
- **AI & Developer Tooling:** Programmed custom Claude / Model Context Protocol (MCP) skills and autonomous sub-agents to automate systemic development workflows, enforcing strict structural QA standards.

Tech Stack: Vue 3 · TypeScript · FastAPI (Python) · Nuxt · Shadcn · Tailwind · Docker · Claude / MCP · Vitest · Playwright · Storybook · Figma

Customer Engineer | Hullabalook Apr 2022 – Jul 2023
London, UK

- **Retail Discovery Apps:** Developed and launched complex client-facing applications via modern CI/CD pipelines utilizing React, Vue, Vuex, and Nuxt.
- **Canvas & UI Tooling:** Built highly visual product discovery interfaces leveraging Fabric.js and the HTML5 Canvas API to create custom interactive customer layouts.
- **Data & Optimization:** Engineered programmatic automated solutions for large, unstructured retail datasets using Python, Google Colab, and GCP to increase data accuracy. Audited web applications to drive substantial performance improvements in Lighthouse and SEO metrics.

Tech Stack: JavaScript · React · Vue · Vuex · Nuxt · Fabric.js · Canvas API · Python · GCP · CI/CD

Architectural Assistant (Part I & II) | Bryden Wood 2015–2016 · 2018–2021
London, UK

- **Systems Thinking & Architecture:** Focused on parametric design, Building Information Modeling (BIM), and algorithmic design paradigms. Specialized in reasoning about vast, interdependent structural data pipelines and platform systems.

FEATURED PRODUCTS & AI RESEARCH

PROFILE

Full-stack engineer with a strong frontend specialization and 4 years of experience shipping production web applications in London. Highly proficient in building reliable, responsive user interfaces with TypeScript, Vue 3, and React, alongside robust backend development in Python (FastAPI). Deeply focused on applied LLM product engineering, automation tooling, and end-to-end feature ownership.

CORE SKILLS

FRONTEND

TypeScript, Vue 3, React 19, Nuxt, Next.js, Tailwind CSS, shadcn/ui, JavaScript

BACKEND & AI

Python, FastAPI, Pydantic, Node.js, RESTful APIs, JWT, Applied LLM Engineering, Claude / MCP, OpenAI SDK, AWS Bedrock

DATA & TESTING

PostgreSQL, Supabase (RLS), Firebase, MongoDB, Vitest, Playwright, Storybook, Figma

INFRA & DEVOPS

AWS (Lambda, SAM), GCP, Docker, Vercel, Git, GitHub Actions (OIDC), CI/CD

CERTIFICATIONS

Claude with Amazon Bedrock

Anthropic / AWS
Issued May 2026

EDUCATION

MSc in Architecture

Politechnika Poznańska
2016 – 2017

BArch in Architecture

Politechnika Poznańska
2011 – 2015

Erasmus Exchange Program

Live: villageos.co.uk | Code: github.com/AdamMrotek/VillageOS

- **System Architecture:** Designed a production-ready family operating system converting multi-modal asynchronous communication (WhatsApp text, PDF newsletters, flyers) into clean, schema-valid calendar events.
- **End-to-End Type Safety:** Established a shared type-contract mapping Pydantic v2 (Python) data models to TypeScript interfaces, eliminating runtime schema-drift between the LLM layer and frontend.
- **Deterministic AI Extraction:** Integrated instructor and Pydantic to force structured LLM outputs. Engineered an isolated golden-dataset evaluation harness tracking per-field extraction accuracy, token latency, and costs.
- **Secure Infrastructure:** Deployed a secure serverless architecture via FastAPI on AWS Lambda (arm64) using AWS SAM. Hardened data isolation using Supabase JWT authorization tied to PostgreSQL Row Level Security (RLS).

Tech Stack: Next.js 16 · React 19 · TypeScript · FastAPI · Pydantic · OpenAI API · Supabase (Postgres RLS) · AWS Lambda / SAM · GitHub Actions

LANGUAGES

English: Full Professional

Polish: Native / Bilingual